

PPE detection on construction imagery — mini report

MAICEN Module 4, Unit 3 · Americo Nuno Caldas Teixeira

Executive summary

We fine-tuned YOLOv8n on a five-class construction PPE dataset to pre-screen site photographs and CCTV frames for people without a hard hat or high-visibility vest. The intended deployment is near real-time triage: a supervisor reviews a ranked worklist instead of sweeping every frame by eye. Because a missed unhelmeted worker is a potential injury and a false alarm costs about thirty seconds, we set a recall-first success criterion.

The model reaches mAP@50 of 0.867, within 1.5 points of the published Roboflow-100 baseline, using a nano backbone and eighteen minutes of training. As a general screening aid it works. It nevertheless **fails the criterion it was built for**: recall on no-helmet is 0.545 against a target of 0.80. Roughly half the unhelmeted people in the validation set pass unflagged. We are reporting a model that is not deployable for its stated safety purpose, and the analysis below explains why that is a data problem rather than a training one.

Results

Class	Precision	Recall	mAP@50	mAP@50–95	Val instances
all	0.863	0.787	0.867	0.474	730
person	0.864	0.919	0.945	0.611	246
helmet	0.921	0.888	0.921	0.522	237
vest	0.818	0.837	0.884	0.495	145
no-vest	0.764	0.746	0.797	0.388	91
no-helmet	0.947	0.545	0.789	0.353	11

Validation split: 121 images, 730 instances. Inference 9.1 ms per image on a T4. Training: yolov8n from COCO weights, 50 epochs configured with patience 15, stopped at epoch 24, best weights from epoch 9, batch 16, imgsz 640, seed 0.

Three readings matter. First, recall tracks training frequency almost exactly — 6,330 helmet instances give 0.888, while 291 no-helmet instances give 0.545. Second, precision on no-helmet is 0.947: when the model fires it is almost always right, it simply requires too much evidence before committing. Third, mAP@50–95 of 0.474 against mAP@50 of 0.867 means boxes fit loosely — acceptable for flagging a frame, unusable for counting people.

What the failure cases actually show

Inspecting the six worst cases changed our reading of the metrics twice over.

The false positives are not errors. All three highest-confidence false positives are correct detections: a white hard hat at 0.91, two people at 0.88 and above. The objects are present; the ground-truth labels are missing or fall below the IoU-0.5 match. Measured precision of 0.863 is therefore a floor rather than a value, and any improvement measured against this validation set is measured against noise. A label audit has to precede further training.

The misses are about context, not frequency. The bare-headed figure we miss most clearly is a woman in a white coat with a stethoscope — no site context, no site dress. The second miss is a man in a soft cap, scored 0.21: below threshold, not absent. Together these suggest no-helmet has learned site context and head-covering shape rather than the absence of a hard hat. Adding more of the same imagery would not fix either case. The dataset is largely web-scraped studio and group photography — one image carries a commercial stock watermark — rather than the site or CCTV frames the deployment assumes.

The third miss is different in kind: a dense scene with twenty-plus people whose heads occupy roughly 30 pixels. That is a scale limit, addressable at inference by tiling large frames or raising `imgsz`, without retraining.

Prioritised improvements

	Action	Targets	Effort
1	Rebalance no-helmet — add 300–500 instances, prioritising caps, hoods and bare heads outside site dress	FN-1, FN-2	~3 h
2	Expand validation support to 50+ no-helmet instances	measurement trust	~1 h
3	Audit the validation labels	FP-1, FP-2	~2 h
4	Add hard negatives — buckets, cones, drums, bystanders	no-vest precision	~2 h

Not on the list: more epochs, a larger backbone, heavier augmentation. Validation mAP plateaued at epoch 9 of 24 — the ceiling is data coverage, not training capacity.

Limitations and governance

This model is an assistive screening tool only. Every detection is reviewed by a site supervisor before any action is taken, and it must never be the sole verifier for a life-safety decision. At current recall roughly half of unhelmeted people pass unflagged; that figure is stated in the repository README rather than buried. Compliance cannot be assessed below about 30 pixels of head height, which is a hard operating limit on camera placement.

Code is MIT licensed. The dataset is CC BY 4.0 with attribution retained and used unmodified; faces are visible in some source images and no client site photography is included. The weights are fine-tuned from Ultralytics YOLOv8 and inherit AGPL-3.0, so distribution or networked deployment carries obligations unless an enterprise licence is obtained. Full detail in `docs/governance_checklist.md`.